Exact Packet Determinants and Sharper Arithmetic–Analytic Budgets for the Claimed Exponent-Two Proof for Pi

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M041

October 8, 2026

Abstract

We examine the complete supplied manuscript claiming that the irrationality exponent of π equals two. This report does not establish that claim or present it as accepted mathematics. For its finite weighted coefficient packets we prove that admissible logarithm truncations preserve every full-row minor exactly. This permits separate arithmetic and analytic truncations. Balanced denominator clearing then removes the dimension and analytic truncation length from the leading arithmetic loss. Exact summation of distinct Taylor degrees, together with the source's available disk, increases its leading quadratic collision constant sixfold. We give explicit finite determinant certificates, a strengthened conditional comparison, and exact executable checks. The center-uniform interpolation theorem, its global geometric inputs, and the unbounded-denominator argument remain separate from these local deductions. Contribution assessment: C3/5, because the report proves substantive source-specific determinant budget improvements but no new irrationality-exponent theorem for pi.

1 Scope and the claim under examination

For irrational $x \in \mathbb{R}$, write

$$\mu(x) = \sup\{\nu > 0 : 0 < |x - p/q| < q^{-\nu} \text{ for infinitely many reduced } p/q,\ q \geq 2\}.$$

The source [1] claims $\mu(\pi) = 2$, and hence convergence of the Flint–Hills series $\sum_{n\geq 1} n^{-3}\sin^{-2} n$ in radians. Its upper-bound statement is eventual and epsilon-dependent; it is not the stronger bounded-partial-quotient assertion $|\pi - p/q| \geq c/q^2$. No result below proves either statement about π.

Applicability is the catalogue's **L3**; the independently assessed contribution is **C3/5**. These are different scales. "Source" denotes an assertion or argument of the supplied manuscript; "local" denotes a theorem proved here for explicitly specified finite or analytic data; "conditional" retains the stated additional premises.

1.1 Actual reading and architecture

All five supplied TeX files were read in bounded, untruncated numbered displays: main 1–264, interpolation 1–584, determinant 1–432, conclusion 1–156, and series 1–155, totaling 1,591 lines. The 269-line bibliography was hashed but not read as a proof source. The two private originals in [6, 7] were read over 380 lines: 129 lines on finite representations, compatible transfer, and polynomial moment unisolvence;

251 lines on local remainders, coefficient derivatives, closure, and invertibility obstructions. Their other passages and the chapter's included subfiles were not read. Exact intervals, complements, and hashes are in the owned read ledger. A full literal manuscript read is not an independent verification of every global theorem it invokes.

Source part	Dependency and review boundary
Introduction	Precise claim, historical bounds, and order of parameters.
Interpolation	Transverse multiplicity; derivative persistence; residue obstruction; curve contact; ordinary blowup; eventual coefficient generation. The complete argument was read; cited intersection and positivity results are not re-proved here.
Determinant	Rational coefficient packet, Gaussian-integer lower bound, exact period translation, repeated-degree cancellation, two alternatives. These are the principal local work sites.
Conclusion	Choose dimension, then approximation denominators, then degree. No finite calculation substitutes for the center-uniform theorem.
Series	Dyadic spacing and nearest-multiple grouping. Convergence remains conditional on the required Diophantine lower bound.

1.2 Current primary literature

The October 8, 2026 refresh distinguishes the published Zeilberger–Zudilin bound $7.103205334137\ldots$ [2] from two recent preprint claims. Bai's September 11 version states $\mu(\pi) < 7.101862832357$ and a two-dimensional *local* optimum of its auxiliary function, not global optimality [3]. Duan's public manuscript states $\mu(\pi) < 7.102252364091047$ and provides exact-certificate code [4]. We inspected their main statements, not their full proofs or certificate implementations. Neither is evidence for exponent two. No confidential title or passage was sent to a public service. The source's own claim is not promoted here to an accepted theorem by the absence of a counterexample in our bounded tests.

2 The interpolation dependency, with quantifiers

Fix $m, K \geq 1$ and rational $w_0, v_0, \theta > 0$ with $0 < \theta < 1$ and

$$K\theta^m < 1, \qquad K(w_0/v_0)\theta^m < 1. \tag{1}$$

The source asserts successive lower thresholds for $w_1, \ldots, w_m$, depending on preceding weights but *not* on the centers. Set

$$W = (w_0, w_1, \ldots, w_m), \qquad V = (v_0, w_1/\theta, \ldots, w_m/\theta), \qquad w\alpha = \sum_{i=1}^{m} w_i\alpha_i.$$

Let $\mathcal{P}_W(H)$ be the polynomials spanned by $Y^h X^\alpha$ with $w_0 h + w\alpha \leq H$. Let $\mathcal{J}_V(H)$ be the formal power-series quotient retaining precisely the monomials $t^s u^\beta$ with $v_0 s + w\beta/\theta < H$. The strict inequality in the target and the weak inequality in the source must not be interchanged. For any centers $(1, c_{j1}, \ldots, c_{jm})$ coordinatewise distinct across $0 \leq j < K$, the asserted map is

$$J_H : P \longmapsto \left(P(1+t, (c_{ji} + u_i + \log(1+t))_{i=1}^m)\right)_{j=0}^{K-1}. \tag{2}$$

Surjectivity is asserted for all sufficiently large divisible H; the divisibility may depend on the weights, and the remaining degree threshold may depend on the centers. Counting dimensions is necessary, not sufficient. A full-rank matrix at one H proves no assertion at arbitrarily larger degrees or at different centers.

2.1 Retained geometric details

The proof is more specific than an unspecified zero estimate. Its commuting frame is $D_0 = Y\partial_Y + \sum_i \partial_{X_i}$, $D_i = \partial_{X_i}$. It preserves the source weighted-degree bound, and in the formal coordinates $D_0 = (1+t)\partial_t$. Persistent derivative vanishing on an algebraic component is compared with isolated intersection length on a transverse coordinate slice. The weighted simplex lower bound is valid without a large-degree threshold. The source explicitly notes that its factor $1/2$ can be omitted; removing that slack is *not* a new contribution of this report.

Successive weight separation compares every pair of generic normal bases. The largest differing positive index yields $dX_i = dY/Y$ on a purported obstruction. Restriction to a complete curve rules out nonconstant Y: an exact meromorphic differential has zero residues, whereas dY/Y has nonzero integer residue at a zero or pole. The fiber $Y = 1$ then uses a second auxiliary polynomial and coordinatewise distinctness. This is an all-curves argument, not a conclusion from a sampled rank.

For the last passage the source uses an ordinary blowup of the ideal generated locally by powers of the regular coordinates. Its branch valuation is Rh_P. The curve inequality makes $A - (1+\sigma)E$ nef; combining it with an ample $aA - E$ makes $A - E$ ample. The sufficiently-high-degree Rees assertions are $p_*\mathcal{O}(-nE) = I^n$ and $R^q p_*\mathcal{O}(-nE) = 0$ for $q > 0$. These use ordinary powers, not an unjustified substitution of integral closures. The statements of the cited Proj lemmas were checked in the primary Stacks source [5]; no small-n equality is asserted. Also, I^n is contained in the weight-$\geq nR$ ideal. Surjectivity onto the larger quotient modulo I^n does imply surjectivity onto the desired smaller packet; equality of ideals is unnecessary.

We have not established a counterexample to this interpolation theorem or a conclusive fatal defect in these displayed steps. Equally, the local results below do not independently certify the global geometric argument. It is retained as an explicit premise whenever a nonzero full-row minor is required asymptotically.

3 Exact coefficient and moment representations

Write $(D_0)_{\underline{s}} = D_0(D_0 - 1)\cdots(D_0 - s + 1)$, with empty product the identity. For the untruncated logarithm packet,

$$[t^s u^\beta]P(1+t, c+u+\log(1+t)) = \frac{1}{s!\beta!}((D_0)_{\underline{s}}\partial_X^\beta P)(1,c). \tag{3}$$

Indeed $\partial_t = Y^{-1}D_0$ along the logarithmic curve, and induction gives $\partial_t^s = Y^{-s}(D_0)_{\underline{s}}$ there. The transverse derivatives commute with D_0. Evaluation at $Y = 1$ proves the formula. Replacing the falling factorial by D_0^s would be wrong: at $s = 2$ the missing term is $-D_0$. Factorials normalize coefficients, not unscaled derivatives.

The private finite representation $m = Ec$, $E_{ij} = \ell_i(\phi_j)$ [6] applies exactly to these functionals on $\mathcal{P}_W(H)$. Here the desired property is *full row rank* and a right inverse for prescribing arbitrary packets, not full column rank and reconstruction from a redundant measurement bank. For a positive-definite coefficient Gram matrix G, full row rank is equivalent to $JG^{-1}J^* > 0$, and

$$R = G^{-1}J^*(JG^{-1}J^*)^{-1}, \qquad JR = I.$$

This elementary representation does not prove its own rank hypothesis. Positive moments are unisolvent on a specified polynomial space when their Gram integral is strictly positive on every nonzero polynomial; positive quadrature weights at too few nodes do not suffice.

The private local-moment Taylor theorem [7] supplies an actual remainder only for its stated integral operator and finite absolute moments. It does not turn arbitrary period or interpolation data into

such an operator. Its variable-coefficient warning is directly relevant to (3): freezing coefficients before differentiating changes the operator. Polynomial exactness is distinct from closure and invertibility; $I - xD$ already kills x while preserving polynomial spaces.

3.1 Admissible changes preserve full minors

At each center choose polynomials $G_i(t)$ with

$$\operatorname{ord}_t(G_i(t) - \log(1+t)) > v_i/v_0, \qquad v_i = w_i/\theta. \tag{4}$$

Let J_H^G denote the packet map with these polynomials in place of the logarithm. The choice may differ between centers.

Theorem 3.1 (Full-minor invariance)**.** *For fixed H, there is a unitriangular automorphism U_G of $\bigoplus_j \mathcal{J}_V(H)$ such that $J_H^G = U_G J_H$. Consequently every square minor using* all *target rows and any fixed equally numerous columns is unchanged. Any two admissible truncation choices give the same such determinant, including when it is zero.*

Proof. Substitute $u_i \mapsto u_i + G_i - \log(1+t)$ and fix t. In the expansion of $t^s u^\beta$, every nonidentity term replaces at least one factor of weight v_i by a term of strictly greater weight. Order the finite packet monomials by nondecreasing weight, with arbitrary order within ties. The induced matrix has diagonal entries one and all other entries strictly on one side of the diagonal. Terms outside the packet cannot return to it. The inverse substitution subtracts the same tails and has the same property. Each center gives a determinant-one block. Thus $\det U_G = 1$; left multiplication of a full-row square submatrix proves the assertion. This is an exact formal identity, not an approximation whose error must be made small. □

For a minor using only a subset of the target rows this conclusion need not hold: the selected row subspace need not be invariant. Nor does the argument cover replacing a low-weight term, as in changing u to $u + ct$ when $v_0 < v_1$. The strict tail condition is the source of invariance.

4 Balanced denominators and independent truncations

Now use the source's arithmetic centers $c_{ji} = jr_i$, $r_i = 2\mathrm{i}p_i/q_i$, with $p_i \in \mathbb{Z}$, $q_i \geq 2$, and $w_i = \lceil \log q_i \rceil$ for $i > 0$. All logarithms are natural. Choose $G_i(t) = \sum_{1 \leq k < T_i} (-1)^{k+1} t^k/k$, where $v_0 T_i > v_i$, and put $L_i = \operatorname{lcm}(1, \ldots, T_i - 1)$, taking $L_i = 1$ when the list is empty. Let Δ be any nonzero full-row minor, with M rows $\rho = (j, s, \beta)$ and selected columns $Y^h X^\alpha$.

Theorem 4.1 (Balanced arithmetic certificate)**.** *Put $a_i = q_i L_i$ and*

$$D_i = \sum_{\text{columns}} \alpha_i - \sum_{\text{rows}} \beta_i.$$

Then $D_i \geq 0$ for every i, and

$$\Delta \prod_i a_i^{D_i} \in \mathbb{Z}[\mathrm{i}] \setminus \{0\}, \qquad \log|\Delta| \geq -\sum_i D_i \log a_i. \tag{5}$$

Define

$$\bar{b}_H = \frac{\sum_\rho w\beta}{MH}, \qquad C_H = \frac{\sum_{\text{columns}} w\alpha}{MH}, \qquad \lambda = \max_i \frac{\log L_i}{w_i}.$$

Then $0 \le C_H - \bar{b}_H \le 1 - \bar{b}_H$, *and*

$$\frac{\log|\Delta|}{MH} \ge -(1+\lambda)(C_H - \bar{b}_H) \ge -(1+\lambda)(1-\bar{b}_H). \tag{6}$$

More precisely $C_H \le 1 - w_0 \sum_{\text{columns}} h/(MH)$.

Proof. Scale column α by $\prod_i a_i^{\alpha_i}$ and row β by $\prod_i a_i^{-\beta_i}$. An entry is zero unless $\alpha \ge \beta$ coordinatewise. Otherwise the scaled entry is

$$\prod_i \binom{\alpha_i}{\beta_i} [t^s](1+t)^h \prod_i (a_i(jr_i + G_i(t)))^{\alpha_i - \beta_i}. \tag{7}$$

Here $a_i jr_i = 2\mathrm{i} j p_i L_i$ and $a_i G_i(t) \in \mathbb{Z}[t]$, so the entry is a Gaussian integer. The determinant is nonzero, giving (5) because a nonzero Gaussian integer has modulus at least one.

There is at least one nonzero permutation product in the original determinant. Every paired entry in that product has $\alpha_i \ge \beta_i$. Summing along the permutation proves $D_i \ge 0$. Since $\log q_i \le w_i$, we have $\log a_i \le (1+\lambda) w_i$ and may multiply this inequality by the nonnegative D_i. Also $\sum_i D_i w_i = MH(C_H - \bar{b}_H)$. Finally the common simplex budget of each column gives $\sum_{\text{columns}} (w_0 h + w\alpha) \le MH$. These observations prove every inequality stated. □

The nonnegativity argument is essential: separately replacing the column and row logarithms by upper bounds would reverse the wrong inequality in the row contribution. Balancing first also removes the source's row ceiling error. This is Gaussian-integer arithmetic, not rounding an approximately integral determinant. Parent-suggested balanced scaling and exponent nonnegativity were independently proved above and tested by the author; parent test totals are not included in this report's totals.

4.1 Decoupling the two logarithm truncations

Use the *short arithmetic truncation*

$$T_i^{\text{ar}} = \left\lfloor \frac{w_i}{\theta v_0} \right\rfloor + 1, \qquad L_i^{\text{ar}} = \operatorname{lcm}\left(1, \ldots, \left\lfloor \frac{w_i}{\theta v_0} \right\rfloor\right). \tag{8}$$

Its omitted term has weight strictly greater than v_i, including when $w_i/(\theta v_0)$ is an integer. For analytic translation use the source's independently chosen

$$T_i^{\text{an}} = \lceil F_0 w_i / v_0 \rceil, \qquad F_0 > 2/\theta.$$

Theorem 3.1 identifies the determinants exactly, so their arithmetic and analytic estimates can be combined.

Corollary 4.2 (Truncation-decoupled lower bound)**.** *With* $\Lambda = 4\log 2$, *put* $\lambda_0 = \Lambda/(\theta v_0)$. *For a nonzero full-row minor of either admissible truncation,*

$$\frac{\log|\Delta|}{MH} \ge -(1+\lambda_0)(1-\bar{b}_H). \tag{9}$$

There is no mF_0/v_0, $\sum_i 1/w_i$, *or row ceiling term in this bound. The sharper exact factors in* (5) *remain available.*

Proof. The elementary estimate $\log \operatorname{lcm}(1,\dots,n) \le \Lambda n$ gives $\log L_i^{\mathrm{ar}} \le \Lambda w_i/(\theta v_0)$. For clarity, if $p^r \in (\ell, 2\ell]$, its prime contributes at least one to the valuation of $\binom{2\ell}{\ell}$ at that scale. Thus the log-LCM increase over the interval is at most $2\ell \log 2$. Sum over dyadic intervals through the least power of two not below n to obtain $4n \log 2$. The empty case is immediate. Apply Theorem 4.1 to the short truncation, then transfer its minor by Theorem 3.1. □

For comparison, the source has additive arithmetic error

$$E_{\mathrm{ar}}^{\mathrm{source}} = \Lambda F_0 m/v_0 + \Lambda \sum_i 1/w_i + \theta/w_*.$$

Our additive error is at most $\lambda_0(1-\bar{b}_H) \le \lambda_0$. Since $F_0 > 2/\theta$, even its leading bound is less than $1/(2m)$ of the source's leading $\Lambda F_0 m/v_0$ term. This is a genuine budget improvement, not a new integrality theorem for π. It relies crucially on complete coefficient packets.

5 Exact collision summation

For a selected monomial column $P = Y^h X^\alpha$, the source uses

$$f_{a,P}(z) = [u^a]P(e^z, z+u) = \begin{cases} \binom{\alpha}{a} e^{hz} z^{|\alpha|-|a|}, & a \le \alpha, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

The same z occurs in all X coordinates. Write $\omega = 2\pi\mathrm{i}$, $\epsilon_i = j(r_i - \omega)$, and $\tau_i = G_i^{\mathrm{an}} - \log(1+t)$. The exact row identity is

$$\mathrm{row}_{j,s,\beta} = \sum_{\substack{a \ge \beta \\ wa \le H}} \binom{a}{\beta} \sum_{0 \le d \le a-\beta} \binom{a-\beta}{d} \epsilon^{a-\beta-d} \sum_{k=0}^{s} [t^k]\tau^d \left([t^{s-k}] f_{a,P}(j\omega + \log(1+t))\right)_P. \tag{11}$$

This follows by expanding the transverse binomial, then the sum of constant displacement and truncation tail. Every scalar is independent of P, which is necessary for determinant multilinearity. The source correctly charges a nonzero tail term by $\sum_i d_i w_i < H/F_0$; a fixed truncation does *not* give an error negligible in every growing-degree packet.

Lemma 5.1 (Distinct-degree partition sum)**.** *For $n \in \mathbb{Z}_{\ge 0}$ and $0 < q < 1$, with empty products equal to one,*

$$\sum_{\substack{d_1,\dots,d_n \ge 0 \\ \text{pairwise distinct}}} q^{d_1+\cdots+d_n} = \frac{n! q^{n(n-1)/2}}{\prod_{k=1}^n (1-q^k)}. \tag{12}$$

Proof. Order the degrees increasingly and write $d_r = r - 1 + e_1 + \cdots + e_r$, $e_r \ge 0$. Their sum is $n(n-1)/2 + \sum_{r=1}^n (n-r+1)e_r$. Sum the resulting independent geometric series and multiply by $n!$ for the orderings. Absolute convergence holds because $0 < q < 1$. □

This is classical antisymmetry/partition algebra, not a new identity. Its quantitative use in this particular determinant is the improvement.

Theorem 5.2 (Sharper uniform collision bound)**.** *Choose one test row in (11) per original row, remove its scalar, and call the resulting M-square matrix T. Let n_a count the rows with the same transverse index a. Put $R = 100K$ and*

$$\mathcal{D}_H = \exp\left\{H\left(R/w_0 + \log(2R)/w_*\right)\right\}, \quad E_{\mathrm{hol}} = R/w_0 + \log(2R)/w_* + \log 2/v_0.$$

For either $q = 1/2$ or $q = 1/8$,

$$|\det T| \le M!\mathcal{D}_H^M 2^{\sum_\rho \ell_\rho} \prod_a \frac{n_a! q^{n_a(n_a-1)/2}}{\prod_{k=1}^{n_a}(1-q^k)}. \tag{13}$$

In particular, with $c_q = \frac{1}{2}\log(1/q)$,

$$\frac{\log|\det T|}{MH} \le -\frac{c_q \sum_a n_a^2}{MH} + E_{\mathrm{hol}} + \rho_H(q), \quad \rho_H(q) = \frac{2\log M + c_q - \log(1-q)}{H}. \tag{14}$$

For fixed parameters $\rho_H(q) \to 0$. At $q = 1/8$ the leading constant is $3\log 2/2$, six times the source's $\log 2/4$.

Proof. On $|z| \le R$, (10) is bounded by $\mathcal{D}_H$: $h \le H/w_0$, $|\alpha| \le H/w_*$ and $\binom{\alpha}{a} \le 2^{|\alpha|}$. Expand $f_{a,P}(z) = \sum_{d\ge 0} c_{a,d,P}(z/R)^d$. Cauchy's estimate gives $|c_{a,d,P}| \le \mathcal{D}_H$. On $|t| = 1/2$,

$$|j\omega + \log(1+t)| \le 2\pi(K-1) + \log 2 < R/8.$$

The last inequality follows already from $\pi < 4$ and $\log 2 < 1$. Thus the test $[t^\ell]$ on $(z/R)^d$ has modulus at most $2^\ell q^d$ for both stated q.

Expand the determinant by the coefficient rows. Absolute convergence follows by bounding each coefficient determinant by $M!\mathcal{D}_H^M$ and summing geometric series. Two equal pairs (a, d) give equal coefficient rows and hence zero. Apply Lemma 5.1 separately in each group to obtain (13). For the uniform form use $\sum n_a = M$, $\prod_a n_a! \le M!$, $\log M! \le M\log M$, and $\prod_a \prod_{k=1}^{n_a}(1-q^k) \ge (1-q)^M$. Also $\sum \ell_\rho \le MH/v_0$. This proves the bound. The source row count is polynomial in H for fixed parameters, so $\log M/H \to 0$ uniformly over all row-term choices and column selections. □

Exact summation at $q = 1/2$ doubles the leading coefficient on its own. Using $q = 1/8$ takes advantage of unused room in the *same* disk, without raising its exponential growth cost. The explicit partition factor is preferable at small degrees; an asymptotic coefficient alone does not show that a simplified finite-H error is smaller in every case.

6 A strengthened conditional comparison

Assume, as in the source determinant section, that fixed integers p_i, q_i satisfy $|\pi - p_i/q_i| \le q_i^{-\nu}$, with $\nu > 2$ and $w_i = \lceil \log q_i \rceil$. No such infinitely large collection is assumed to exist as a fact. Keep $F_0 > 2/\theta$ for analytic translation and set

$$\begin{aligned} E_{\mathrm{tr}} &= \nu/F_0 + \log 2/v_0 + (\log 4 + \log(2K) + \nu)/w_*, \\ E &= E_{\mathrm{tr}} + E_{\mathrm{hol}}, \qquad Q_H = (\lfloor H \rfloor + 1)^{2m}(\lfloor H/v_0 \rfloor + 1), \\ \varepsilon_H &= \rho_H(1/8) + \log Q_H/H, \qquad c = 3\log 2/2. \end{aligned}$$

The bound on a scalar in (11) is $\exp\{-\nu w(a-\beta) + HE_{\mathrm{tr}}\}$: the approximation errors give $-\nu w(a-\beta-d)$, a contributing tail has $wd < H/F_0$, Cauchy's estimate contributes 2^s, and the two binomials contribute at most $4^{|a|}$. The ceiling inequality $q_i^{-\nu} \le e^\nu e^{-\nu w_i}$ is retained here; it is not needed in the balanced arithmetic bound.

For $0 < A, \eta < 1$, define exact finite counts

$$D_A(H) = \#\{a \in \mathbb{Z}_{\ge 0}^m : wa \le AH\}, \qquad L_H = \frac{\eta^2 M_H}{HD_A(H)}.$$

The fixed-parameter simplex counts give

$$\bar{b}_H \longrightarrow b_* := \frac{m\theta}{m+2}, \qquad L_H \longrightarrow L := \frac{\eta^2 K \theta^m}{(m+1)v_0 A^m}. \tag{15}$$

To verify the first limit, scale the row simplex by H and change variables to $x_0 = v_0 s/H$, $x_i = w_i \beta_i/(\theta H)$. The uniform simplex in dimension $m+1$ has mean $1/(m+2)$ in each coordinate: integrating one coordinate gives this value directly, or symmetry among the $m+2$ barycentric coordinates does. The weighted transverse sum has mean $m\theta/(m+2)$. Polynomial Riemann sums, with a boundary of measure zero, justify the limit. This sharpens the source's bound $\bar{b}_H \le \theta$, without changing its order of limits.

Theorem 6.1 (Conditional finite and subsequence tests)**.** *Under the fixed data above, a nonzero full-row minor at degree H is impossible if both inequalities hold:*

$$\begin{aligned} cL_H &> (1+\lambda_0)(1-\bar{b}_H) + E + \varepsilon_H, \\ \nu(A(1-\eta) - \bar{b}_H) &> (1+\lambda_0)(1-\bar{b}_H) + E + \varepsilon_H. \end{aligned} \tag{16}$$

Consequently, nonzero full-row minors on an unbounded subsequence *of admissible H are already incompatible with*

$$\begin{aligned} cL &> (1+\lambda_0)(1-b_*) + E, \\ \nu(A(1-\eta) - b_*) &> (1+\lambda_0)(1-b_*) + E. \end{aligned} \tag{17}$$

The geometric claim of full surjectivity in every sufficiently large degree is sufficient but stronger than the rank premise used here.

Proof. There are at most Q_H^M choices of translated row terms. If at least ηM selected indices have $wa \le AH$, then $\sum_a n_a^2 \ge (\eta M)^2/D_A(H) = MHL_H$ by Cauchy–Schwarz. The remaining scalar main exponent is nonpositive because $a \ge \beta$. Otherwise more than $(1-\eta)M$ indices have $wa > AH$, and the scalar exponent is at most $-\nu MH(A(1-\eta) - \bar{b}_H)$. Discard the nonpositive collision term only in this second case. Theorem 5.2, absolute summation and the term count yield

$$\frac{\log|\Delta|}{MH} \le E + \varepsilon_H + \max\{-cL_H, -\nu(A(1-\eta) - \bar{b}_H)\}.$$

The determinant here uses the long analytic truncation. Its equality with the short arithmetic one permits Corollary 4.2. The two strict inequalities contradict that lower bound. Finally (15) and $\varepsilon_H \to 0$ show that (17) forces (16) for all sufficiently large H, hence also on any unbounded subsequence. □

These improvements decrease a local error budget and weaken the *degree* quantifier needed by the determinant comparison. They do not establish the remaining center-uniform rank premise, an effective threshold, or an infinite set of exceptional approximations. The source chooses $\theta, A, B, C, \eta, F_0$, then a finite m with $K = \lfloor C^m \rfloor$, $w_0 = B^{-m}$, $v_0 = 2K\theta^m w_0$; only then are sufficiently separated q_i selected, and only afterwards does $H \to \infty$. Our theorem respects that order. Allowing weights or centers to vary with H would invalidate the stated error limits without new uniform estimates.

7 Finite analytic certificates and what they certify

The representation in Section 3 supports exact rational coefficient extraction, fraction-free elimination, or a modular rank witness on an integral scaled matrix. A nonzero determinant modulo a Gaussian

prime is sufficient for nonvanishing over $\mathbb{Q}(\mathrm{i})$; a zero residue at one prime does not prove singularity. All scaling factors, row indices and selected columns must be retained in the certificate.

For a Taylor cutoff N, truncate each expansion in the proof of Theorem 5.2 at degree $d = N$. Its entrywise remainder at a test row of order ℓ is at most

$$\mathcal{D}_H 2^{\ell} \frac{q^{N+1}}{1-q}. \tag{18}$$

Thus the row ℓ^1 error is at most M times this bound, before adding any independent coefficient or evaluation enclosure errors. The truncated quantities still involve π at exact period centers; floating-point evaluation does not make them exact rational data.

Proposition 7.1 (Determinant enclosure)**.** *Let $Z = B + F$ be an M-square complex matrix. Suppose the ℓ^1 norms of the rows of B and F are at most b_i, e_i, respectively. Then*

$$|\det Z - \det B| \le \prod_{i=1}^{M}(b_i + e_i) - \prod_{i=1}^{M} b_i =: \mathcal{E}. \tag{19}$$

If, in addition, exact arithmetic establishes $\det Z \in \delta\mathbb{Z}[\mathrm{i}]$ with $\delta > 0$, then $|\det B| + \mathcal{E} < \delta$ implies $\det Z = 0$. Conversely $|\det B| > \mathcal{E}$ implies $\det Z \neq 0$ without the lattice premise.

Proof. Expand the determinant by row multilinearity and remove the term with every row from B. Bound each remaining determinant by the product of its row ℓ^2 norms, hence by its row ℓ^1 norms. Summing over the nonempty subsets gives (19). A nonzero member of $\delta\mathbb{Z}[\mathrm{i}]$ has modulus at least δ. The direct and reverse triangle inequalities give the two conclusions. □

For our exact minor the scale is $\delta = \prod_i (q_i L_i^{\mathrm{ar}})^{-D_i}$. This certificate proves a statement about that *one* determinant. It does not certify a nonzero minor at every large degree. Conversely, if exact elimination supplies a nonzero minor, an analytic upper enclosure strictly below its lattice gap exposes incompatible premises or an implementation error; it is not an instruction to round the nonzero determinant to zero.

Relation to Padé constructions. Prescribing zero coefficients is a finite Hermite–Padé-type linear constraint. Neither a small residual nor exact finite solvability provides an irrationality exponent by itself. An arithmetic linear form requires exact coefficient denominators, nonvanishing, suitable size estimates and an index-selection mechanism reaching arbitrary rational denominators. The source instead obtains a contradiction from several hypothetical exceptional approximations via a geometric interpolation theorem. Our finite packet machinery does not replace that theorem by a numerical approximation to π.

8 Retained series consequences and limitations

The source's series argument has a valid conditional spacing mechanism. If irrational $\alpha > 0$ satisfies $\|q\alpha\| \ge cq^{1-\nu}$ for all integers $q \ge 1$, with $c > 0$ and $1 < \nu < 5/2$, then in $K \le q < 2K$ the circle points are separated from each other and from zero by $d = c(2K)^{1-\nu}$. On each half-circle the ordered distances are at least $d, 2d, \ldots$, so

$$\sum_{K \le q < 2K} q^{-3} \|q\alpha\|^{-2} \le 2K^{-3} d^{-2} \sum_{j \ge 1} j^{-2} = O(K^{2\nu-5}).$$

The dyadic sum converges. For $\alpha = \pi$, group integer n by a nearest q to n/π. The group has at most four integers, $n \ge \pi q/2$, and $|\sin n| \ge (2/\pi)\|q\pi\|$. This gives the source's factor $8/\pi$ and proves

convergence *under the Diophantine premise*. A pointwise sine bound without the separation count would miss the relevant saving.

Likewise the claimed criterion $a > \max(1, b)$ for $\sum n^{-a}|\sin n|^{-b}$ uses exponent-two input for its convergence half. The divergence when $a \le 1$ follows from $|\sin n| \le 1$; for $1 < a \le b$, convergent numerators p_j have $|\sin p_j| < 1/q_j$ and give summands bounded away from zero. These retained observations do not independently settle Flint–Hills, nor the boundary at irrationality exponent $5/2$ discussed by the source.

9 Assessment, corrections, and next steps

No source file was edited. No conclusive mathematical error was established in the supplied proof during this scoped investigation. The report therefore does not invent a correction or certify the claimed resolution. The following distinctions matter:

- The source already proves filtration preservation; our further deduction is determinant-one action on the complete packet and the resulting independent choices of arithmetic and analytic truncation.
- The source's denominator estimate is valid but wasteful. Balanced scaling, aggregate nonnegative exponent differences, and the shortest admissible arithmetic truncation give the improved bound proved here.
- The source's half-decay collision bound is valid. Exact distinct-degree summation and its available radius sharpen the constant, not the global rank theorem. The partition identity itself is classical.
- The multiplicity factor two, strict target cutoff, possibly center-dependent large-degree threshold, ordinary-power Rees recovery, Gaussian-integer modulus gap, and full-simplex budgets are retained accurately. Their removal or reinterpretation would create errors.

Advantages and costs. Short arithmetic truncations reduce denominator sizes, while long analytic truncations retain strong error suppression. Complete packets make the minor invariant, so no determinant perturbation is paid for this switch. The collision improvement keeps the source's growth radius fixed. Exact finite counts and the selected columns can sharpen the comparison further. Costs remain substantial: dimensions grow like H^{m+1}, separated weights can be enormous, and exact elimination can have severe coefficient growth. Complete-packet invariance is unavailable for arbitrary sparse row selections. A small-degree experiment supplies no uniform Seshadri-type or interpolation bound.

Feasible future work. An implementable next step is a certificate generator that records the packet, a modularly witnessed full minor, the balanced exponent vector, both truncations and outward-rounded analytic bounds. Column selection could minimize the exact cost $\sum_i D_i \log(q_i L_i)$ instead of merely finding a pivot minor. A separate geometric project would need an independently checked, preferably effective center-uniform interpolation theorem. Those tasks should not be confused: improving a determinant's budget cannot discharge an unproved global rank premise.

Evidence and score. The owned exact-check program uses manufactured rational/Gaussian-rational packets, factorial-normalized differential rows, admissible truncation changes, balanced denominators, distinct-degree sums and finite determinant error bounds. Its recorded count refers only to assertions actually run by this author. Parent checks are separately attributed; neither set is a test of $\mu(\pi) = 2$. Full-page visual inspection and compilation check the artifact, not the theorem. On the five-point

contribution scale, C1 is no demonstrated increment, C2 limited, C3 notable local, C4 substantial and C5 very high. We assign C3 for the proved truncation-decoupled arithmetic and sharpened collision budgets, not for renaming moment coordinates or for the source's global claim. No historical-priority claim is made. C4 or C5 would require new global arithmetic or geometric information absent from this report.

References

[1] OpenAI, *The irrationality exponent of pi is 2*, September 24, 2026. Pinned public source. The owned ledger identifies the five TeX files and hashes.

[2] D. Zeilberger and W. Zudilin, *The irrationality measure of π is at most* 7.103205334137..., Moscow J. Combin. Number Theory 9 (2020), 407–419. arXiv:1912.06345v2.

[3] Y. Bai, *The irrationality measure of π is at most* 7.101862832357, preprint, arXiv:2609.11276v2, September 11, 2026. Main statements inspected, full proof not audited.

[4] J. Duan, *A refinement of the Zeilberger–Zudilin integral and an improved upper bound for the irrationality measure of pi*, 2026. Public manuscript and code repository. Main statement inspected; code not executed or audited here.

[5] The Stacks Project, Lemmas 30.14.2 and 30.14.3, Tag 0AG6 and Tag 0AG7. Primary statements and displayed proofs inspected; their dependencies imported.

[6] M. Subbarao, private original `TR-rev-1-old-tr250.tex`, selected passages on finite Ram representations, compatible coordinate transfer and polynomial moment unisolvence. Exact read intervals in the local ledger.

[7] M. Subbarao, private original `ch03-local-moments-rev2.tex`, selected actual proofs on local moment remainders, coefficient differentiation and degree closure. Included files are outside the read scope.